Afonso Teodoro – AI Engineer, PhD

Location	Lisbon, Portugal	**Email**	afonso.teodoro91@gmail.com
Work	Remote, or relocation	**Web**	amteodoro.github.io
Nationality	Portuguese (EU)	**LinkedIn**	linkedin.com/in/afonso-teodoro
Languages	Portuguese, English (C2)	**GitHub**	github.com/amteodoro

Summary

Seven years shipping production machine learning for insurance and healthcare clients — document understanding at ∼150k documents a month, vehicle-damage detection, and fine-tuned speech-to-text — owning components from annotation tooling and models through APIs and deployment. PhD in inverse imaging problems (Instituto Superior Técnico, 19/20), IBM Scientific Prize 2019, 400+ citations. Open to senior, lead or founding AI engineering roles — remote, Lisbon or relocation; also available for select consulting engagements.

Experience

Jun 2019 - Present Nimble Portal
Artificial Intelligence Consultant

Owned components of client-facing computer vision, document-understanding and speech systems for insurance and healthcare operations: models, annotation and review tooling, inference APIs, containerised deployment and handover.

- Helped build the document-understanding pipeline for medical-expense reimbursement: ∼**150k invoices per month**, with **70%+ of reimbursements settled without human review**. PyTorch models behind FastAPI, deployed with Docker; designed the human-review path for the remaining cases.
- Vehicle-damage detection service used in underwriting and fraud-prevention workflows: trained the detection models, built the React.js annotation platform the labelling operation ran on, shipped the inference service and an Android proof-of-concept for in-field capture.
- Fine-tuned Whisper for domain-specific transcription in an insurance call-centre pilot; added speaker diarization and LLM summarisation over the transcripts for call analytics.

Additional engagement: consulting on a medical-imaging data-science platform covering 2D/3D volume labelling through to integrated model training.

Jun 2015 - May 2019 Instituto de Telecomunicações
PhD Researcher

Machine learning and optimisation methods for inverse imaging problems — deblurring, super-resolution, MRI reconstruction and hyperspectral image fusion for remote sensing. First-author papers in IEEE TIP, TCI and TGRS (400+ citations); released reference implementations publicly in MATLAB, C++ and Python.

Dec 2014 - May 2015 Instituto de Telecomunicações
Research Fellow

Project PAConvex — extended the Gaussian-mixture image denoiser from my MSc thesis.

Feb 2013 - Jun 2013 INESC-ID
Research Intern
Project ARGUS — built a trajectory-monitoring and abnormal-event detection interface.

Selected Projects

– **Portugal Fire Detection** — live demo · source (MIT)
End-to-end satellite fire-detection service. Ingests NASA FIRMS, Meteosat and Sentinel-3 feeds, runs a four-channel MobileViT classifier over Sentinel-2 tiles, and alerts on user-drawn areas of interest. FastAPI, PostGIS, async SQLAlchemy, Alembic, APScheduler, rate limiting, Docker Compose.

– **Nomly** — nomly.xyz
Installable web app (PWA) for gluten-free living: barcode scanning against product databases, plus an AI model that reads unfamiliar ingredient labels from a photo and flags hidden gluten sources. Restaurant discovery and pantry tracking. Designed, built and shipped solo.

Education

2015-2019 PhD, Electrical and Computer Engineering — Instituto Superior Técnico, Lisbon
19/20. Thesis: *Mixture Models for Inverse Imaging Problems — Going Beyond Denoising in a Plug-and-Play Fashion.*
Supervisor: Prof. Mário Figueiredo.

2012-2014 MSc, Electrical and Computer Engineering — Instituto Superior Técnico, Lisbon
18/20. Thesis: *Single-frame Image Denoising and Inpainting using Gaussian Mixture Models.*

Awards

– IBM Scientific Prize 2019 — "Adapt to survive: targeted priors in plug-and-play image restoration".

Selected Publications

1. A. Teodoro, J. Bioucas-Dias, M. Figueiredo, "A Convergent Image Fusion Algorithm Using Scene-Adapted Gaussian-Mixture-Based Denoising", *IEEE Trans. Image Processing*, 28(1), 2019. 105 citations.
2. A. Teodoro, J. Bioucas-Dias, M. Figueiredo, "Image Restoration and Reconstruction Using Variable Splitting and Class-Adapted Image Priors", *IEEE ICIP*, 2016. 96 citations.
3. A. Teodoro, J. Bioucas-Dias, M. Figueiredo, "Image Restoration Using Targeted Plug-and-Play Priors", *IEEE Trans. Computational Imaging*, 5(4), 2019. 40 citations.

Full list, 400+ citations: Google Scholar.

Skills

Machine & deep learning: *Python, PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Hugging Face Transformers. Computer vision: detection, segmentation, OCR and document understanding, remote sensing. NLP: information extraction, text classification. Speech: Whisper fine-tuning, diarization.*

LLM systems: *RAG, embeddings and retrieval, evaluation pipelines, prompt design; LangChain, OpenAI / Anthropic / OpenRouter APIs.*

Engineering: *FastAPI, Docker and Compose, PostgreSQL/PostGIS, SQLAlchemy, Alembic, Git, CI/CD, Cloudflare Workers, Hugging Face Spaces; React.js, JavaScript, HTML/CSS; MATLAB, C++.*

Updated November 2026 · current version always at amteodoro.github.io/cv_afonso.pdf